A CLOSED SUBSCHEME IS AN EFFECTIVE CARTIER DIVISOR IF AND ONLY IF IT IS LOCALLY CUT OUT BY A SINGLE NONZERODIVISOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-characterize-effective-Cartier-divisor is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 2402-2413 and proof span 2415-2433 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. This is the largest complete proof among the zero-external-reference candidates of the divisors halfA pool, and the only one of them that is not an author-declared omission.

Editorial note. The two directions are the author's own; the phrase 'In other words' at proof line 2418 is kept verbatim, so the identification of the generator with an element of A stays the author's step and was not rewritten.

Allowed standard prerequisites (editorial).

- Effective Cartier divisors via the ideal sheaf: a closed subscheme D of a scheme S with ideal sheaf I_D is an effective Cartier divisor exactly when I_D is an invertible O_S-module, and on the open complement S minus D the restriction of I_D is O. Supplies both directions of the equivalence used and concluded at proof lines 2415-2418 and 2430-2432. Gives the target of the second direction and the starting point of the first: invertibility of I_D is what has to be translated into the local equation f.
- Quasi-coherence of ideal sheaves and the tilde construction over an affine open Spec(A): sections of I_D over U correspond to an ideal of A, and D cap U = Spec(A/(f)) when the restriction of I_D is generated by f. Used at proof lines 2418-2421 and again at 2426-2429. Converts the chosen free generator f of the sections of I_D into an element of A and identifies the closed subscheme it cuts out.

Lemma 0.1. *Let S be a scheme. Let $D \subset S$ be a closed subscheme. The following are equivalent:*

(1) *The subscheme D is an effective Cartier divisor on S.*

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(2) *For every $x \in D$ there exists an affine open neighbourhood $\mathrm{Spec}(A) = U \subset S$ of x such that $U \cap D = \mathrm{Spec}(A/(f))$ with $f \in A$ a nonzerodivisor.*

Proof. Assume (1). For every $x \in D$ there exists an affine open neighbourhood $\mathrm{Spec}(A) = U \subset S$ of x such that $\mathcal{I}_D|_U \cong \mathcal{O}_U$. In other words, there exists a section $f \in \Gamma(U, \mathcal{I}_D)$ which freely generates the restriction $\mathcal{I}_D|_U$. Hence $f \in A$, and the multiplication map $f : A \to A$ is injective. Also, since $\mathcal{I}_D$ is quasi-coherent we see that $D \cap U = \mathrm{Spec}(A/(f))$.

Assume (2). Let $x \in D$. By assumption there exists an affine open neighbourhood $\mathrm{Spec}(A) = U \subset S$ of x such that $U \cap D = \mathrm{Spec}(A/(f))$ with $f \in A$ a nonzerodivisor. Then $\mathcal{I}_D|_U \cong \mathcal{O}_U$ since it is equal to $\widetilde{(f)} \cong \widetilde{A} \cong \mathcal{O}_U$. Of course $\mathcal{I}_D$ restricted to the open subscheme $S \setminus D$ is isomorphic to $\mathcal{O}_{S \setminus D}$. Hence $\mathcal{I}_D$ is an invertible $\mathcal{O}_S$-module. $\square$